

Lecture 25 (Jan 4, 2026): Learning from experts & Multiplicative weight update

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In this lecture we start a new algorithm design technique called **Multiplicative Weights Algorithms** or online learning or learning from experts.

25.1 The expert Problem (or prediction with expert advice)

Suppose we have n experts that make predictions for something. e.g. market goes up or down, It is going to rain or not, etc.

- Let U be the universe of possible outcomes and at the start of day t each expert $i \in [n]$ makes a prediction $p_i^t \in U$.
- The algorithm receives all these predictions & must come up with its own prediction $o^t \in U$.
- We then are given the actual outcome at t . The goal is to minimize the mistakes we make i.e. # of days our prediction was false.

We obviously cannot do better than the best expert. Our goal is to be as close to the best expert in hindsight: After, Say, T number of days, We want our mistakes not (much) larger than that of the expert with min # of mistakes in those T days.

Formally, given

- m : # of mistakes by our alg.
- m_i : # mistakes of expert $i \in [n]$
- $m^* = \min m_i$

Goal: minimize the difference $(m - m^*)$.

Lemma 1 *There is an algorithm that makes $\log n$ mistakes if there is a perfect expert.*

Proof. Suppose $m^* = 0$, i.e. there is an expert that makes no mistake.

- Each day let o^t be the majority of predictions.
- At the end of each day remove experts that were wrong.

Note that for each day we make a mistake the # of remaining experts goes down by a factor of ≥ 2 (because the correct answer was not majority) $\Rightarrow$ # of mistakes $\leq \lceil \log n \rceil$, ■

Now consider the general case (no perfect expert). We can generalize:

Lemma 2 *There is an algorithm that makes at most $(m^* + 1) \log n$ mistakes.*

Proof.

- Think of time divided into "epochs"
- We run the algorithm of "perfect expert" until experts are removed.
- Bring back all experts & Start a new epoch.

Suppose $m^* = m_{i^*}$. Note that in each epoch we make at most $\lceil \log n \rceil$ mistakes. Also # of epochs $\leq m^*$: each time all experts are eliminated $\equiv i^*$ made a mistake. Finally in the last epoch (after m^*) there are at most $\lceil \log n \rceil$ mistakes. ■

25.2 The Weighted Majority Algorithm (WM)

We can improve by softening our penalty: instead of eliminating an expert who erred, discount their weight! [Littlestone & Warmuth '89]

- For each expert i maintain weight w_i^t for each day t and let $w_i^1 = 1$.
- Each day $t \geq 1$, return weighted majority

$$o^t = \arg \max_{u \in U} \sum_{i: i \text{ predicts } u} w_i^t$$

- After seeing the outcome for each wrong expert Set: $w_i^{t+1} = w_i^t / 2$

Theorem 1 *For any sequence of predictions, the number of mistakes made by WM is $\leq 2.41(m^* + \log n)$.*

Proof. We show the # of mistakes $\leq 2.41(m_i + \log n) \quad \forall i$. Define potential function

$$\Phi^t = \sum_{1 \leq i \leq n} w_i^t$$

Note:

1. $\Phi^1 = n$,
2. $\Phi^{t+1} \leq \Phi^t \quad \forall t$
3. If WM algorithm makes a mistake in round t then total weight of wrong experts is more than correct ones:

$$\begin{aligned} \Phi^{t+1} &= \sum_{i \text{ wrong}} w_i^{t+1} + \sum_{i \text{ correct}} w_i^{t+1} \\ &= \frac{1}{2} \sum_{i \text{ wrong}} w_i^t + \sum_{i \text{ correct}} w_i^t \\ &= \Phi^t - \frac{1}{2} \sum_{i \text{ wrong}} w_i^t \\ &\leq \frac{3}{4} \Phi^t \quad (\text{Since weight of wrong experts is } > \frac{1}{2} \Phi^t) \end{aligned}$$

If after T rounds, i has made m_i mistakes and WM has made m mistakes then

$$\begin{aligned} \left(\frac{1}{2}\right)^{m_i} &= w_i^{T+1} \leq \Phi^{T+1} \leq \Phi^1 \left(\frac{3}{4}\right)^m = n \left(\frac{3}{4}\right)^m \\ \Rightarrow m &\leq \frac{m_i + \log n}{\log \frac{4}{3}} \leq 2.41(m_i + \log n). \end{aligned}$$

■

Using a similar analysis and choosing $\epsilon \in (0, \frac{1}{2})$:

Theorem 2 For $\epsilon \in (0, \frac{1}{2})$, Penalizing each wrong expert by a factor of $(1 - \epsilon)$ guarantees the # of mistakes by WM to be

$$m \leq 2(1 + \epsilon)m^* + O\left(\frac{\log n}{\epsilon}\right)$$

Proof. Same analysis implies $\Phi^{t+1} \leq (1 - \frac{\epsilon}{2})\Phi^t$ and

$$\begin{aligned} (1 - \epsilon)^{m_i} &\leq \Phi^{T+1} \leq \Phi^1 \left(1 - \frac{\epsilon}{2}\right)^m = n \left(1 - \frac{\epsilon}{2}\right)^m \\ \Rightarrow m &\leq \frac{-m_i \log(1 - \epsilon) + \ln n}{\epsilon/2} \leq \frac{2m_i(\epsilon + \epsilon^2)}{\epsilon} + O\left(\frac{\log n}{\epsilon}\right). \\ &\text{(using the fact that } -\ln(1 - \epsilon) = \epsilon + \frac{\epsilon^2}{2} + \frac{\epsilon^3}{3} + \dots \leq \epsilon + \epsilon^2 \dots) \end{aligned}$$

This means the ratio gets close to 2. ■

Proposition 1 Any deterministic algorithm makes $\geq 2m^*$ mistakes on some input.

Proof. If the algorithm is deterministic adversary knows the prediction based on history & prediction of experts. Suppose $n = 2$:

- A always predicts 1,
- B always predicts 0.

So the adversary can choose the outcome s.t. the algorithm is always wrong but at least one of A or B has $\leq \frac{1}{2}$ error rate! ■

25.3 The Multiplicative Weight update (randomized WM)

Now we see the randomized version of Weighted majority that is known as the MWU algorithm. We use randomization to bypass the issue of scenario in the prev. Proposition.

Intuition: Use the weights exactly as before but now the prediction is drawn randomly proportional to current weights.

$$Pr[o^t = u] = \frac{\sum_{i:p_i^t=u} w_i^t}{\sum_i w_i^t}$$

The update of w_i^t 's is the same as before.

Theorem 3 For any $\epsilon \leq \frac{1}{2}$, the expected number of mistakes made by randomized WM is:

$$E[m] \leq (1 + \epsilon)m_i + O\left(\frac{\log n}{\epsilon}\right)$$

Proof. Again define potential function

$$\Phi^t = \sum_i w_i^t$$

Let

$$F^t = \frac{\sum_{i \text{ wrong}} w_i^t}{\sum_i w_i^t}$$

Note: $E[m] = \sum_{t \in [T]} F^t$ as we make a mistake with probability F^t at each step t .

$$\begin{aligned} \Phi^{t+1} &= \sum_{i \text{ wrong}} w_i^{t+1} + \sum_{i \text{ correct}} w_i^{t+1} \\ &= \sum_{i \text{ correct}} w_i^t + \sum_{i \text{ wrong}} (1 - \epsilon)w_i^t \\ &= \Phi^t((1 - F_t) + (1 - \epsilon)F_t) = \Phi^t(1 - \epsilon F_t) \end{aligned}$$

Therefore, we have

$$\begin{aligned} (1 - \epsilon)^{m_i} &\leq \Phi^{T+1} = \Phi^1 \prod_{t=1}^T (1 - \epsilon F_t) \leq n e^{-\epsilon \sum_t F_t} = n e^{-\epsilon E[m]} \\ &\Rightarrow m_i \log(1 - \epsilon) \leq \ln n - \epsilon E[m] \\ &\Rightarrow E[m] \leq (1 + \epsilon)m_i + O\left(\frac{\log n}{\epsilon}\right). \end{aligned}$$

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